MIL-S-12109C
31 October 1968
SUPERSEDING
MIL-S-12109B
17 April 1959

MILITARY SPECIFICATION

SOLENOID, ELECTRICAL; GUN-FIRING MECHANISMS

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers solenoids for use in gun-firing mechanisms used by the Military Services (see 6.1).

1.2 Classification. Solenoids shall be furnished in the following classes, as specified on the applicable specification sheet (see 6.2).

Class 1 - Solenoids for cannons.
Class 2 - Solenoids for machine guns.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal
L-P-509 - Plastic Sheet, Rod and Tube, Laminated Thermosetting.
QQ-P-416- Plating Cadmium (Electrodeposited).

Military
MIL-M-14 - Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-C-13486 - Cable, Special Purpose, Electrical: Low Tension, Heavy Duty, Single-Conductor and Multiconductor.
MIL-T-13513 - Terminals, Lug; Electrical, Solderless Splices, Conductor: Electrical, Solderless (For Automotive use).
MIL-E-13856 - Electrical Components for Automotive Vehicles; Waterproofness Test.

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STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Indentification Marking of US Military Property.
MIL-STD-193	- Painting Procedures, Tactical Vehicles (Tracked and Wheeled).
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-810	- Enviromental Test Methods.
MS 33800	- Connector, Receptacle, Electrical-Pin Contact, No. 12, 14, and 16 A.W.G., Waterproof.

(Copies of specifications, standards, drawings, and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following doucment forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Supertintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Qualification. The solenoids furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.3).

3.2 Design and construction. Solenoids shall be constructed to the form and dimensions specified on the applicable Military Specification Sheet listed on the Supplement to this specification (see 6.2) and shall be grounded internally.

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3.2.1 Interchangeability. To provide for interchangeability, solenoids fabricated in accordance with this specification shall have external dimensions that fall within the envelope outline of the applicable specification sheet. Electrical connectors, shaft and mounting dimensions and location shall also conform to the detail requirements of the applicable document.

3.2.2 Materials. Materials shall be as specified herein, on applicable standards or drawings, and in applicable specification sheets.

3.2.2.1 Dissimilar metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals, as defined in requirement 16 of MIL-STD-454; which would encourage galvanic action, shall be avoided. Separation of dissimilar metals shall be accomplished by providing insulation between mating surfaces.

3.2.3 Clearance. Clearance between any current-carrying part and other conducting parts of different potential shall be no less than 1/16 inch through air or insulation.

3.2.4 Standard parts. Military standard parts shall be incorporated wherever applicable. Commercial standard parts may be used provided they are interchangeable with Military Standard parts without modification, and the equivalent MS part is identified by the manufacturer.

3.2.5 Threaded parts. All screw threads shall conform to National Bureau of Standards Handbook H28.

3.2.6 Springs. Springs shall not be used to transmit current.

3.2.7 Locking devices. Lockwashers, self-locking nuts, or other approved locking devices shall be incorporated where specified or where required to prevent loosening of components.

3.2.8 Solder. Solder and soldering flux shall conform to requirement 5, of MIL-STD-454.

3.2.9 Plastic materials.

3.2.9.1 Molded. Molded insulation materials shall conform to type MFE requirements of MIL-M-14.

3.2.9.2 Laminated. Laminated plastic materials shall conform to type IV, Grade G-5 or G-6 requirements of L-P-509.

3.2.10 Terminals. Terminals shall conform to MS33800.

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3.2.11 Cable. Cable shall conform to MIL-C-13486.

3.2.12 Plunger springs. Plunger return springs shall be provided on class 2 solenoids.

3.2.13 Crimping. Crimping of conductors shall conform to MIL-T-13513.

3.3 Performance.

3.3.1 Functional (operating). When tested as specified in 4.6.1, the solenoids shall operate over a voltage range of 18 to 30 volts DC.

3.3.2 Current. When measured as specified in 4.6.2, the current shall not exceed the values specified on the applicable specification sheet.

3.3.3 Static force. When tested as specified in 4.6.3, the static force shall be in accordance with the applicable specification sheet.

3.3.4 Firing (class 2 solenoids only). After testing as specified in 4.6.4, class 2 solenoids shall meet the requirements of 3.3.1.

3.3.5 Continuous duty (class 2 solenoids only). While testing as specified in 4.6.5, class 2 solenoids shall meet the static force on the applicable specification sheet.

3.3.6 Endurance. Throughout the test specified in 4.6.6, the solenoids shall produce the static force specified in 3.3.3.

3.3.7 Terminal strength. When tested as specified in 4.6.7, the cable shall evidence no permanent deformation exceeding 1/16 inch, insulation rupture, or failure to meet any requirement of this specification.

3.4 Environmental.

3.4.1 Waterproofness. When tested as specified in 4.7.1, there shall be no leakage into the coil winding or into the termination of the cable to the coil winding. For type II, class 2 components of MIL-E-13856, the solenoids shall then meet the requirements of 3.3.1.

3.4.2 Salt Fog. After testing as specified in 4.7.2, the solenoids shall meet the requirements of 3.3.1.

3.4.3 Fungus. After testing as specified in 4.7.3, the solenoids shall meet the requirements of 3.3.1.

3.4.4 Vibration. After testing as specified in 4.7.4, the solenoids shall meet the requirements of 3.3.1.

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3.4.5 Shock. After testing as specified in 4.7.5, the solenoids shall meet the requirements of 3.3.1.

3.4.6 Sand and dust resistance. After testing as specified in 4.7.6, the solenoids shall meet the requirements of 3.3.1.

3.4.7 Temperature extremes. When tested as specified in 4.7.7, the solenoids shall meet the requirements of 3.3.1.

3.5 Finish. The solenoids shall be finished as specified in 3.5.1 through 3.5.3.

3.5.1 Painting. Cleaning, pretreatment, priming, and painting shall conform to the requirements of MIL-STD-193 (see 6.2).

3.5.2 Electroplating. Exposed, unprotected metal surfaces shall be electroplated in accordance with class 2, type II of QQ-P-416.

3.5.3 Color. Exterior surface color shall conform to colors specified in MIL-STD-193 (see 6.2).

3.6 Marking.

3.6.1 Identification marking. Identification marking shall be in accordance with MIL-STD-130 and as a minimum shall include the following:

Military Part Number (see 6.2)
Manufacturer's Name or Identification
Manufacturer's Serial Number or
Date of Manufacture (Month, Year e.g. FEB 67)
U S

3.7 Workmanship. Workmanship shall be such as to assure a product free of burrs, rust, scratches, and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification testing. A qualification sample of eight solenoids, of each class to be tested, shall be submitted for qualification testing. The samples shall be representative of the units proposed to be furnished under the

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contract. Qualification testing shall be conducted by the Government at a place designated by the Government and shall consist of examination for the defects specified in 4.3.2.2 and testing as specified in table I in the order listed.

Table I. Order of qualification testing.

Specimen No.	Paragraph No.	Test sequence
	4.6.7	Terminal strength
	4.6.2	Current
	4.6.1	Functional (operating)
	4.6.3	Static force
	4.7.4	Vibration
1 and 2	4.6.1	Functional (operating)
	4.7.5	Shock
	4.6.2	Current
	4.6.1	Functional (operating)
	4.6.3	Static force
	4.7.7	Temperature extremes
	4.6.7	Terminal strength
	4.6.2	Current
	4.6.1	Functional (operating)
	4.6.3	Static force
3 and 4	4.7.6	Sand and dust resistance
	4.6.1	Functional (operating)
	4.6.6	Endurance
	4.6.2	Current
	4.6.1	Functional (operating)
	4.6.3	Static force
	4.6.7	Terminal strength
	4.6.2	Current
	4.6.1	Functional (operating)
	4.6.3	Static force
5 and 6	4.7.3	Fungus
	4.7.1	Waterproofness
	4.7.2	Salt Fog
	4.6.2	Current
	4.6.1	Functional (operating)
	4.6.3	Static force
	4.6.7	Terminal strength
7 and 8	4.6.4	Firing (class 2 only)

4.2.1 Failure. Failure of any specimen to pass any examination or test specified herein may be cause, at the option of the Government, for refusal to

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conduct additional testing until the faults revealed by the test have been corrected.

4.3 Quality conformance inspection.

4.3.1 Sampling.

4.3.1.1 Lot formation. A lot shall consist of all solenoids of one class, of one part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.3.1.2 Sampling for examination. Samples for quality conformance shall be selected in accordance with MIL-STD-105.

4.3.1.3 Sampling for testing. A representative sample shall be selected in accordance with level S-3 of MIL-STD-105 from each lot that has passed the inspection specified in 4.3.2.2.

4.3.2 Examination.

4.3.2.1 Acceptable quality level. Each solenoid selected in accordance with 4.3.1.2 shall be examined for conformance to the following acceptable quality levels (AQL) on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

4.3.2.2 Classification of defects. For examination purposes, defects shall be classified as specified in table II.

Table II. Classification of defects.

Categories		Defects	Method of inspection
Major	101	Dimensions not within tolerances specified on specification sheet (see 3.2)	Gage and Visual
Minor	201	Identification marking (see 3.6)	Visual
	202	Workmanship (see 3.7)	Visual

4.3.3 Testing.

4.3.3.1 Quality conformance test. Samples selected in accordance with 4.3.1.3 shall be subjected to the tests specified in table III using an AQL of 6.5 on the basis of percent defective.

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Table III. Order of quality conformance testing.

Paragraph	Test
4.6.2	Current
4.6.1	Functional (operating)
4.6.5	Continuous duty (class 2 solenoids only)

4.4 Control test.

4.4.1 Sampling for control test. Solenoids shall be selected at the rate of three of each 500 produced or fraction thereof, providing that not more than six units shall be selected in any 30-day period. The control test sample shall be identified as to production period and after having been examined for the defects specified in 4.3.2.2, they shall be subjected to the tests specified in table IV in the order listed.

Table IV. Order of control testing.

Sample number	Paragraph	Test
1 and 4	4.6.5	Continuous duty (class 2 solenoids only)
2 and 5	4.6.6	Endurance
3 and 6	4.7.4	Vibration

4.4.2 Failure. Failure of any control test sample to pass any examination or test specified herein may be cause, at the option of the Government, for refusal to accept subsequent lots until the faults revealed by the test have been corrected.

4.5 Test procedures.

4.5.1 General.

4.5.1.1 Conditions. Tests shall be conducted under general laboratory conditions acceptable to the Government. Unless otherwise specified herein, the ambient air (room) temperature shall be $77^{\circ} \pm 15^{\circ}$F. Specimens shall be temperature stabilized at the applicable ambient temperature for one hour before being subjected to tests. However, soaking need not be repeated where tests are continuous at that temperature.

4.5.1.2 Apparatus. In addition to standard measuring equipment (see table V and applicable test specifications), test apparatus shall include a testing chamber capable of maintaining specified air temperatures over the range from

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minus 60°F to plus 130°F. Air movement in the high temperature chamber shall be held to a minimum to prevent a temperature gradient within the chamber and any cooling resulting from excess air motion. There shall also be a submersion tank capable of producing and maintaining a pressure of six pounds per square inch (psi) and a vacuum of six psi.

4.5.1.3 Calibration of test equipment. Unless specified herein, test equipment accuracy shall be such as to permit measurement of 10 percent of product or test specification tolerance. Calibration of test equipment shall be conducted at intervals sufficient to establish required accuracy. Records of calibration shall be made available to the Government. The inspector may refuse to conduct an inspection where accuracy of test equipment has not been established to the satisfaction of the Government. Required measuring instrument tolerances are as listed in table V.

Table V. Measuring instrument tolerances.

Measuring instrument	Accuracy
Voltmeter DC	± 0.50 percent
Ammeter DC (multi-range)	± 0.50 percent

4.6 Performance test.

4.6.1 Functional (operating). To determine conformance to 3.3.1, the solenoid shall be mounted on a test stand and tested with no load at 18 and 30 volts DC. Solenoids shall be cycled as specified in 4.6.6 for 45 minutes at each voltage.

4.6.2 Current. To determine conformance to 3.3.2, the current shall be measured before the test specified in 4.6.1.

4.6.3 Static force. To determine conformance to 3.3.3, the static force shall be measured before and immediately after the test specified in 4.6.1.

4.6.4 Firing (class 2 solenoids only). To determine conformance to 3.3.4, solenoid shall be installed on the gun for which it is intended and 1,000 rounds of ammunition fired by cycling the solenoid four seconds on and eight seconds off. Applied voltage shall be 18 volts DC.

4.6.5 Continuous duty (class 2 solenoids only). To determine conformance to 3.3.5, solenoid shall be mounted as in intended operation and continuously energized for 90 seconds. The applied voltage shall be 30 volts DC; ambient air temperature shall be 125 ± 5°F.

4.6.6 Endurance. To determine conformance to 3.3.6, solenoids shall be tested as follows:

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4.6.6.1 Class 1. Class 1 solenoid shall be operated for a total of 10,000 cycles, each cycle consisting of 0.60 second on and 10 seconds off, while operating against the static load specified in 3.3.3; applied voltage shall be 28 volts DC.

4.6.6.2 Class 2. Class 2 solenoid shall be tested as specified in 4.6.6.1, except that each cycle shall consist of four seconds on and eight seconds off.

4.6.7 Terminal strength. To determine conformance to 3.3.7, a force of 45 pounds shall be exerted between the cable and the terminal along their mutual centerline for one minute. The cable shall be subjected to a pull of 25 pounds along, at 45°, and perpendicular to the centerline of the solenoid for one minute (see figure 1).

4.7 Environmental test.

4.7.1 Waterproofness. The solenoids shall be tested as specified for type II, class 2 components of MIL-E-13856. There shall be no leakage into the coil winding or into the termination of the coil winding. Water entering into the plunger or armature area or the escape of air bubbles from this area during the waterproofness test shall not be considered as leakage. The solenoid shall then be subjected to tests specified in 4.7.2, 4.6.2, 4.6.1, and 4.6.3 to determine conformance to 3.4.1.

4.7.2 Salt Fog. To determine conformance to 3.4.2, the solenoids shall be subjected to the salt fog test specified in method 509 of MIL-STD-810, except that the duration of the test shall be 200 hours.

4.7.3 Fungus. To determine conformance to 3.4.3, the solenoids shall be subjected to the fungus test specified in method 508 of MIL-STD-810, except that the test duration shall be 90 days.

4.7.4 Vibration. To determine conformance to 3.4.4, the solenoids shall be subjected to the vibration test specified in MIL-STD-810, method 514, procedure I, parts 1, 2 and 3, figure 5 (except the maximum load shall be 10G). One of the vibration axes shall be through or parallel to the solenoid plunger.

4.7.5 Shock. To determine conformance to 3.4.5, the solenoids shall be subjected to the shock test in method 516, procedure I, figure 1, of MIL-STD-810. One of the shock axes shall be through or parallel to the solenoid plunger.

4.7.6 Sand and dust resistance. To determine conformance to 3.4.6, the solenoids shall be subjected to the sand and dust test specified in test condition A, method 110 of MIL-STD-202. The velocity shall be 2,300 ± 500 feet per minute.

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4.7.7 Temperature extremes. To determine conformance to 3.4.7, the solenoids shall be tested as follows:

4.7.7.1 Low temperature. The solenoid shall be mounted on a base, placed in a temperature test chamber, the electrical circuits connected as in intended use, at minus 65° ± 5°F for 24 hours. After 24 hour period, and while at this temperature, solenoid shall be subjected to the test specified in 4.6.3.

4.7.7.2 High temperature. The solenoid shall be allowed to return to a temperature of 77° ± 15°F and again tested as specified in 4.7.7.1, except that the ambient air temperature shall be 125° ± 5°F.

4.8 Inspection of preparation for delivery.

4.8.1 Materials and processes. The Government inspector shall, at unscheduled intervals, inspect all materials and processes involved in the preparation for delivery, to determine conformance to the requirements of section 5. Any evidence of deviation from specified requirements shall be cause, at the option of the Government, for refusal to conduct further inspection until objective evidence has been provided by the contractor that corrective action has been taken.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking, for the desired level of protection, shall be in accordance with the applicable packaging standard or packaging data sheet specified by the procuring activity (see 6.2).

6. NOTES

6.1 Intended use. Solenoids covered by this specification are intended for use in the operation of gun-firing mechanisms but may be used for other purposes where the characteristics herein specified are applicable.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Part number (see applicable specification sheet).
(c) Selection of applicable level and packaging standard or packaging data sheet (see 5.1).
(d) Finish (see 3.5).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Product List whether or not such products have actually been so listed by that date. The attention

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of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is U.S. Army Tank-Automotive Command, Warren, Michigan 48090 and information pertaining to qualification of products may be obtained from that activity.

Custodians:

Army - AT

Navy - OS

Air Force - 85

Preparing activity:

Army - AT

Project No. 5945-0144

Review Activities:

Army - MU, WC

User Activities:

Army - AV

Navy - AS, MC

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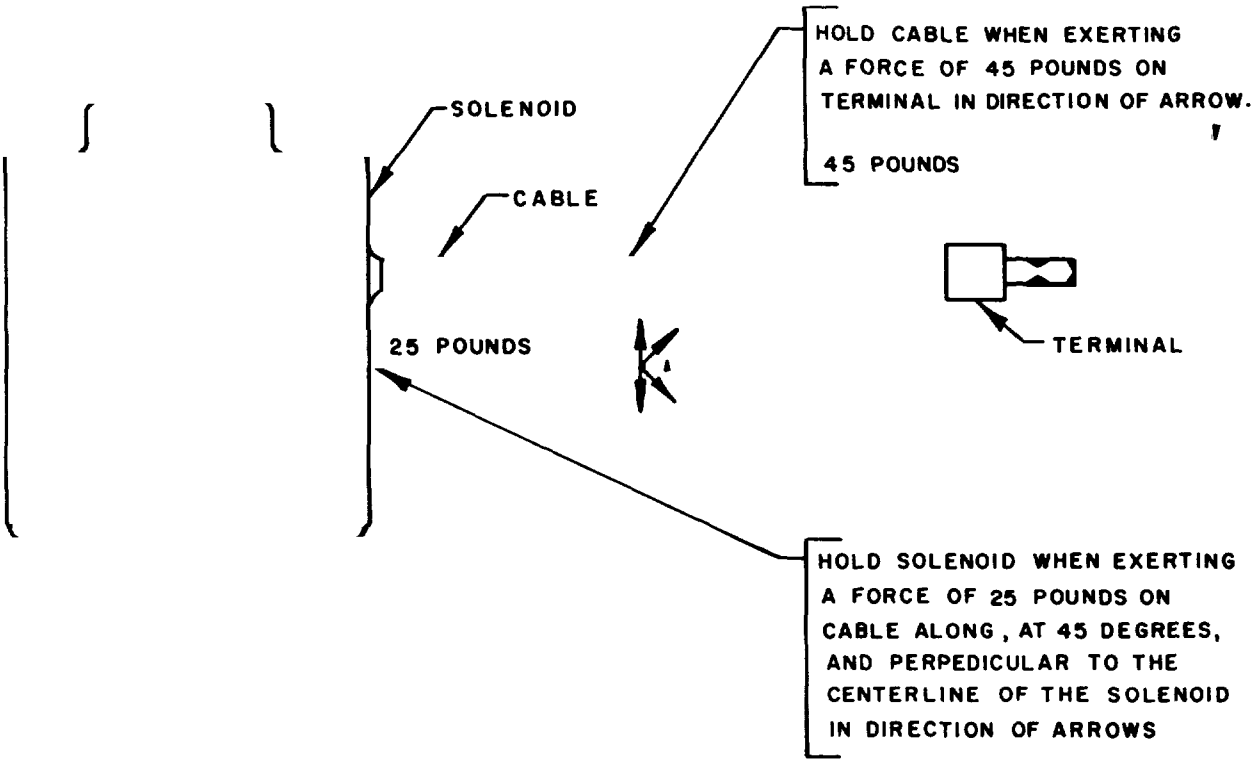
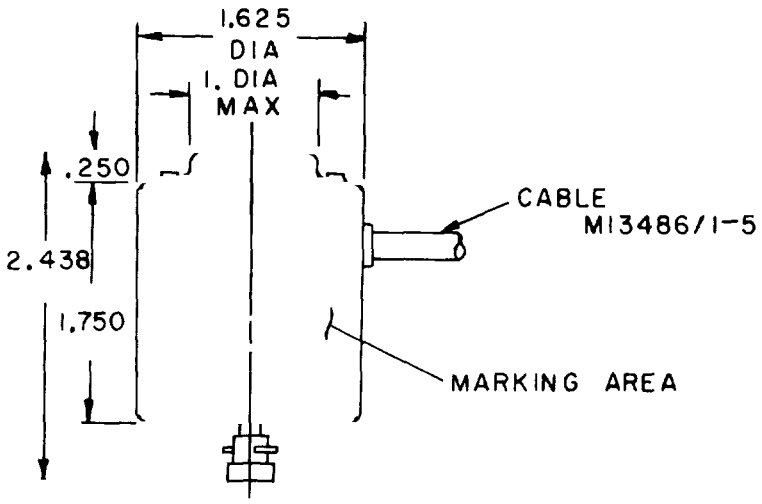


FIGURE 1. TERMINAL STRENGTH TEST

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NOTE: TOLERANCES ARE ± 010 ON DECIMAL

SOLENOID, ELECTRICAL, CLASS 2
(MACHINE GUN), CAL 30, 24 VOLT DC
4.5 AMP MAX

FIGURE 2

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel, either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION

MIL-S-12109C

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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